# C++

BY: JANHAVI DEO

**Functions in C++ :**

A function is a group of statements that is executed when it is called from some point of the program.

Functions are used to structure our programs in a more modular way.

Gives feature of reusability because of its write once and call multiple times mechanism.

They are written to perform a specific task and are also called as module or sub-program or procedure or sub-routine.

A function can optionally define input parameters that enable callers to pass arguments into the function.

A function can optionally return a value as output.

Every function should be called directly or indirectly from main(). main() function is entry point function for C++ program.

type function_name ( parameter1, parameter2, ...) { statements }

where:

- type is the data type specifier of the data returned by the function.
- name is the identifier by which it will be possible to call the function.
- parameters (as many as needed): Each parameter consists of a data type specifier followed by an identifier, like any regular variable declaration (for example: int x) and which acts within the function as a regular local variable. They allow to pass arguments to the function when it is called. The different parameters are separated by commas.

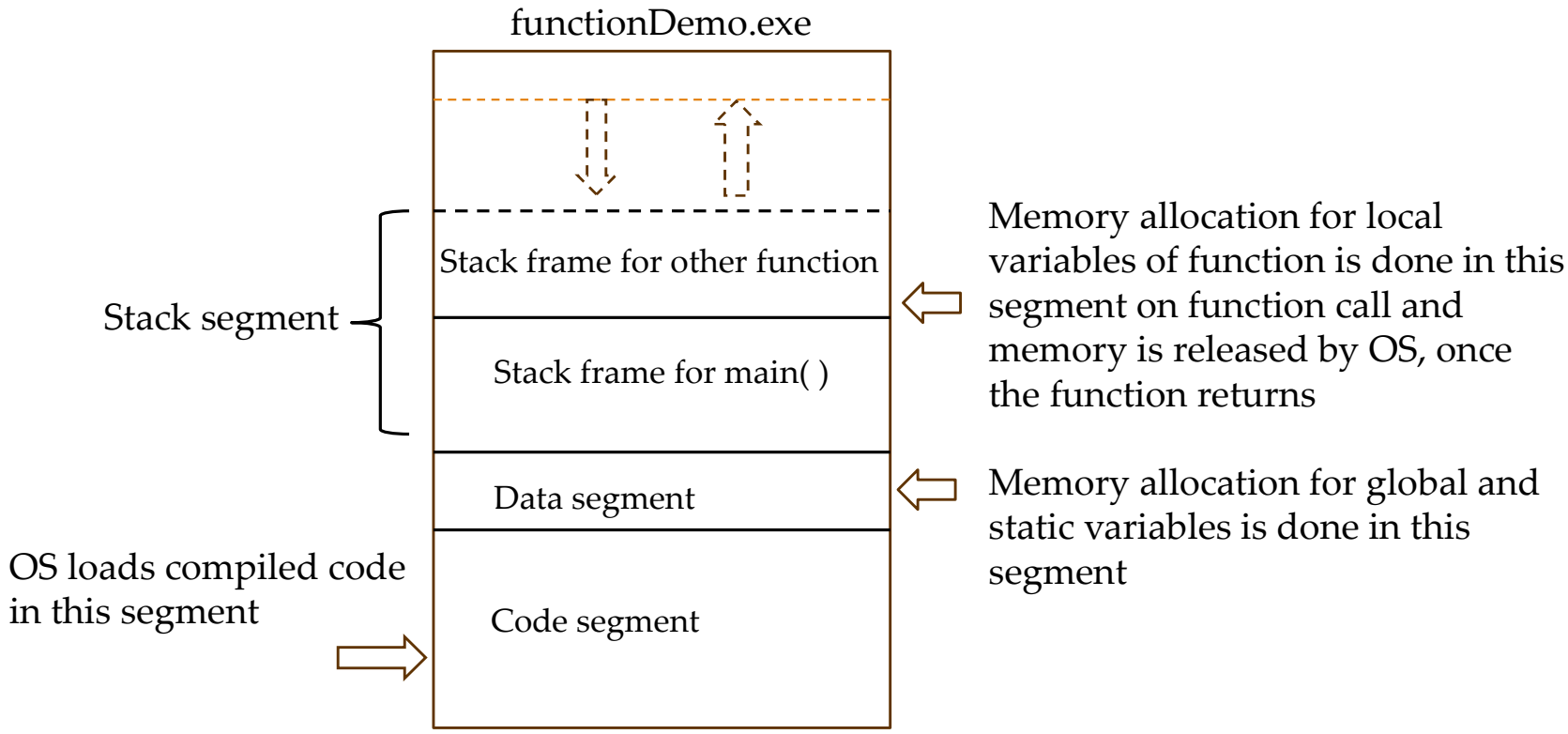
- statements is the function's body. It is a block of statements surrounded by braces { }.

Different forms of functions:

- Function returning nothing and taking no arguments.
- Function return a value and taking no arguments.
- Function returning nothing and taking arguments.
- Function returning a value and taking arguments.

Function declaration, definition and call should match to avoid syntax error.

**Understand function call with respect to memory allocation done by Operating System**



**Sample code to understand function**

```
#include<iostream>
using namespace std;

void display_hello();        // function declaration

int main()
{
  cout<<"\n Welcome";
  display_hello() ;          // function call
  cout<<"\n The end...";
  return 0;
}

void display_hello()         //function definition
{
  cout<<"\n Hello";
  cout<<"\n World....";
}
```

```
#include<iostream>
using namespace std;
void print_num(int num, int );  //function declaration

int main( )
{
  int n;
  cout<<"\n Enter a number..";
  cin>>n;
  print_num(n, n+45);           //function call
  cout<<"\nThe end...";
  return 0;
}
```

Actual Parameters

Formal Parameters

```
void print_num(int x, int y)  //function definition
{
cout<<"\n Printing number thru function..."<<x;
  cout<<"\n Printing 2nd argument.."<<y;
 }
```

***** The value of actual parameters are assigned to formal parameters.**

***** The number of actual and formal parameters should be same**

***** The datatypes of actual and formal parameters should match**

***** Actual parameter can be literals or variables or expression, formal should be variables**

## Sample code to understand **function**

```
#include<iostream>
using namespace std;

// function declaration
int add(int,int);
int sub(int,int);
int mul(int,int);
int division(int,int);
int modulo(int,int);

int main()
{
  int num1, num2;
  cout<<"\n Enter 2 numbers..";
  cin>>num1;
  cin>>num2;
```

```
cout<<"\n Addition  = "<<add(num1,10);
cout<<"\n Subtraction = "<<sub(num1,num2);
cout<<"\n Multiplication = "<<mul(num1, num2);
cout<<"\n Division = "<<division(num1, num2);
cout<<"\n Modulo = "<<modulo(num1, num2);
return 0;
}// end of main
```

```
int add(int x, int y)
{
   int res;
   res = x + y;
   return res;
}
```

```
int sub(int a, int b)
{
  return a - b;
}
```

# Default Parameter in function

**Default Parameters** feature of C++, allows us to assign a value to a function parameter so that if the caller doesn't provide one, the default is used.

```
#include<iostream>
using namespace std;
int sum(int a, int b = 20)
{
   return a + b;
}
/*int sum(int a) //need to discard this definition
{
   return a+900;
}*/
```

```
int main()
{
   int num1= 10;
   cout<<"\n sum = "<<sum(num1);
   return 0;
}
```

**Assignments:**

1. Write functions for add, subtract, multiply and divide two numbers. Write menu driven program to input two numbers from user and call appropriate function based on choice given by user.

2. Write menu driven program to find out
    1. Factorial
    2. Sum of digits
    3. Reverse a number
    4. Print prime numbers for a range
    5. Square of a number
    6. Exit.

    Write appropriate functions and call functions.

3. Write menu driven program for 1. isPrime 2. IsEven 3. IsOdd 4. IsDigit 5. IsPositive 6. Exit

**More Assignments:**

1. Implement pow(x, n), which calculates x raised to the power n (i.e., xn).

2. Write a function int max_of_four(int a, int b, int c, int d) which returns the maximum of the four arguments it receives.

3. A number n is said to be special if the sum of its digits is divisible by 4.

4. Write a function greet_person(…) with two default parameter, one is of type string (person_name) and second is int (n), to greet the person n times. Default parameter values are “Guest” and 1